

View Factor for a Differential Area and a Finite Rectangle in Parallel Planes

Question

To illustrate the numerical integration of Equation,

$$F_{d_{1-2}} = \frac{1}{\pi} \int_{y_{2a}}^{y_{2b}} I_{x2}(y_2) dy_2,$$

consider the two sets of data given in Table 1. In the first case, we shall obtain the view factors shown in the last row of the table. In the second case, we shall obtain a plot of the view factor for data Set 1 as a function of the separation distance of the surfaces for $0.1 \leq Z_o \leq 5$.

View Factors for Two Plate Configurations and View Factor Results			
Parameter		Set 1	Set 2
Geometry of A2			
X coordinate of first corner point, x_{2a}		-1	-1
Y coordinate of first corner point, y_{2a}		-1	-1
X coordinate of opposite corner point, x_{2b}		0	1
Y coordinate of opposite corner point, y_{2b}		0	1
Separation distance between planes, z_o		5	1
Computed view factor $F_{d_{1-2}}$		0.0121	0.5541

Table 1 View Factors for Two Plate Configurations and View Factor Results

Create two subfunctions **Fd1_2** and **kernel2**, which are used to determine the kernel $f(x_2, y_2)$ at any location on surface A_2 . The function **kernel2** uses a vector-based formulation to determine the length S and $\cos \theta$. It is noted that for compatibility with `dblquad`, **kernel2** must return a vector whose length equals the length of the input vector. This allows `dblquad` to minimize the number of calls to the integrand function while still providing the needed data. This capability is implemented in **kernel2** by using `length` to determine the number of elements in x . Use these functions to evaluate the two data sets given in Table 1.

Solution

1. The Analytical Formula

The code numerically solves the following integral for the view factor ($F_{d_{1-2}}$) from a differential area dA_1 at the origin to a parallel rectangle A_2 :

$$F_{d1-2} = \frac{1}{\pi} \int_{y_a}^{y_b} \int_{x_a}^{x_b} \frac{dz^2}{(x^2 + y^2 + dz^2)^2} dx dy$$

For a rectangle with one corner at the origin and extending to coordinates (L_1, L_2) at a separation distance D , the formula is:

$$F_0 = \frac{1}{2\pi} \left[\frac{X}{\sqrt{1+X^2}} \tan^{-1} \left(\frac{Y}{\sqrt{1+X^2}} \right) + \frac{Y}{\sqrt{1+Y^2}} \tan^{-1} \left(\frac{X}{\sqrt{1+Y^2}} \right) \right]$$

where $X = L_1/D$ and $Y = L_2/D$ are dimensionless lengths.

2. View Factor Algebra (Superposition)

To handle rectangles not at the origin, use the principle of superposition. The view factor to a complex shape can be found by adding or subtracting the view factors of simpler shapes that share a common corner with the differential area.

For a rectangle defined by corners (x_a, y_a) and (x_b, y_b) , the total view factor can be assembled from four “base case” rectangles:

$$F_{total} = F_0(x_b, y_b) - F_0(x_a, y_b) - F_0(x_b, y_a) + F_0(x_a, y_a)$$

3. Hand Calculation for Case 1

The code calculates $F_{d1-2}(-1, 0, -1, 0, 5)$. This is a rectangle in the third quadrant.

- **Geometry:** A rectangle with corners at $(-1, -1)$ and $(0, 0)$. By symmetry, its view factor is identical to a rectangle in the first quadrant with corners at $(0, 0)$ and $(1, 1)$.
- **Parameters:**
 - ✓ $L_1 = 1$
 - ✓ $L_2 = 1$
 - ✓ $D = dz = 5$
- **Dimensionless Ratios:**
 - ✓ $X = L_1/D = 1/5 = 0.2$
 - ✓ $Y = L_2/D = 1/5 = 0.2$
- **Calculation:**

Since $X = Y$, the two terms in the analytical formula are identical.

$$F_{d1-2} = \frac{1}{2\pi} \left[2 \cdot \frac{X}{\sqrt{1+X^2}} \tan^{-1} \left(\frac{Y}{\sqrt{1+X^2}} \right) \right]$$

$$F_{d1-2} = \frac{1}{\pi} \left[\frac{0.2}{\sqrt{1+0.2^2}} \tan^{-1} \left(\frac{0.2}{\sqrt{1+0.2^2}} \right) \right]$$

$$F_{d1-2} = \frac{1}{\pi} \left[\frac{0.2}{\sqrt{1.04}} \tan^{-1} \left(\frac{0.2}{\sqrt{1.04}} \right) \right]$$

$$F_{d1-2} = \frac{1}{\pi} \left[\frac{0.2}{1.0198} \tan^{-1} \left(\frac{0.2}{1.0198} \right) \right]$$

$$F_{d1-2} = \frac{1}{\pi} [0.1961 \cdot \tan^{-1}(0.1961)]$$

Using radians for the arctan: $\tan^{-1}(0.1961) \approx 0.1942 \text{ rad}$

$$F_{d1-2} = \frac{1}{\pi} [0.1961 \cdot 0.1961] = \frac{0.03808}{\pi}$$

$$F_{d1-2} \approx 0.01212$$

4. Hand Calculation for Case 2

The code calculates $F_{d1_2}(-1, 1, -1, 1, 1)$. This is a larger rectangle centered on the origin.

- **Geometry:** We can treat this as four identical quadrants. We'll calculate the view factor to the first quadrant (corners at (0,0) and (1,1)) and multiply the result by 4.

- **Parameters (for one quadrant):**

$$\checkmark L_1 = 1$$

$$\checkmark L_2 = 1$$

$$\checkmark D = d_z = 1$$

- **Dimensionless Ratios:**

$$\checkmark X = L_1/D = 1/1 = 1$$

$$\checkmark Y = L_2/D = 1/1 = 1$$

- **Calculation (for one quadrant):**

Again, since $X = Y$, the formula simplifies.

$$F_{\text{quadrant}} = \frac{1}{\pi} \left[\frac{X}{\sqrt{1+X^2}} \tan^{-1} \left(\frac{Y}{\sqrt{1+X^2}} \right) \right]$$

$$F_{\text{quadrant}} = \frac{1}{\pi} \left[\frac{1}{\sqrt{1+1^2}} \tan^{-1} \left(\frac{1}{\sqrt{1+1^2}} \right) \right]$$

$$F_{\text{quadrant}} = \frac{1}{\pi} \left[\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{1}{\sqrt{2}} \right) \right]$$

$$F_{\text{quadrant}} = \frac{1}{\pi} [0.7071 \cdot \tan^{-1}(0.7071)]$$

Using radians: $\tan^{-1}(0.7071) \approx 0.6155 \text{ rad}$

$$F_{\text{quadrant}} = \frac{1}{\pi} [0.7071 \cdot 0.6155] = \frac{0.4352}{\pi}$$

$$F_{\text{quadrant}} \approx 0.13856$$

- **Total View Factor:**

The total view factor is 4 times the view factor of one quadrant.

$$F_{d1-2} = 4 \cdot F_{\text{quadrant}} = 4 \cdot 0.13856$$

$$F_{d1-2} \approx \mathbf{0.5542}$$

Result

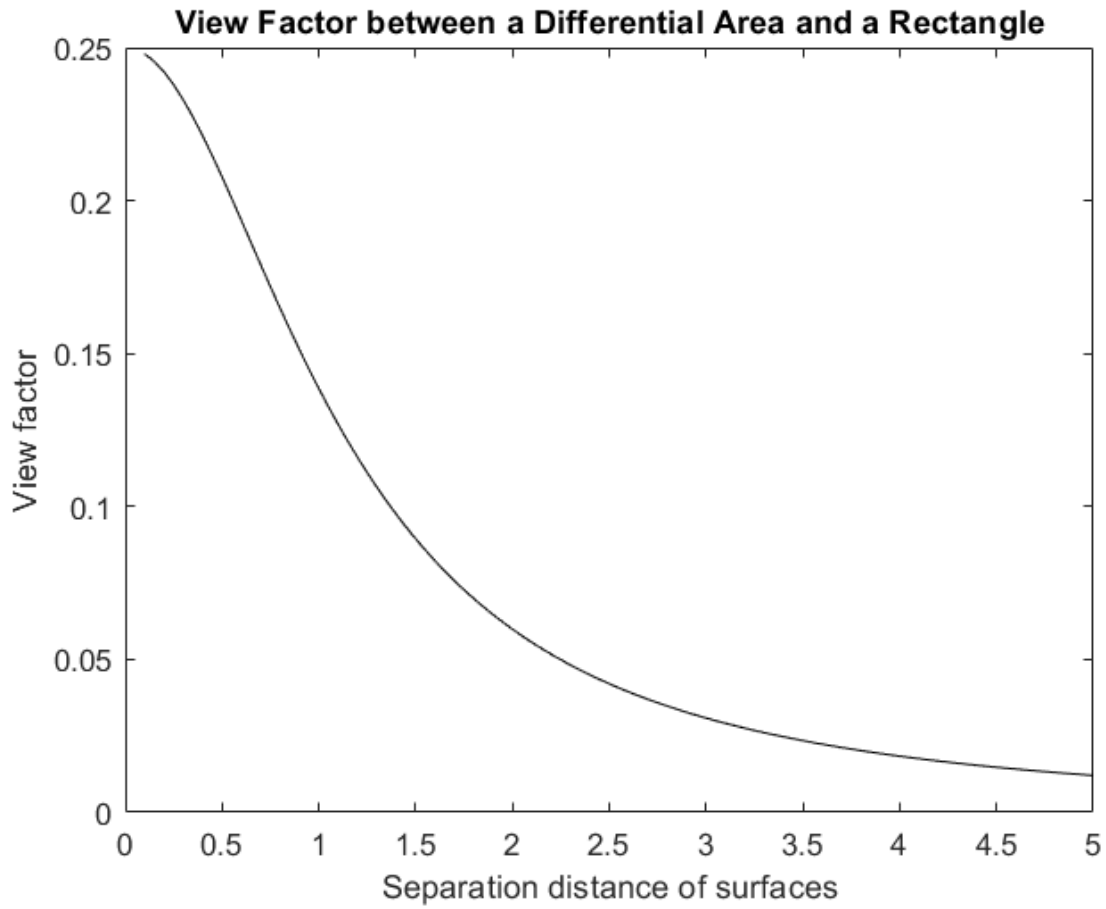


Figure 1 View factor versus separation distance between two parallel planes for the geometry. The areas are in parallel planes and the differential area is aligned with a corner of the finite area.